

JEDI-LETKF Experiments (Version 2)

This version allows users to run a nature run, generate observations, generate an initial background ensemble and run cycling LETKF experiments. This gives users more control over designing their own experiments.

Step-by-step instructions are provided below.

Step 1: Log in to RACC2

```
ssh -X USERNAME@racc.rdg.ac.uk
```

Step 2: Set up the experiment folder

```
cp /storage/research/nceo/DA-training-course/docs/setup_letkf_exps_v2.sh .
```

```
bash setup_letkf_exps_v2.sh
```

```
cd letkf_exps
```

Step 3: Run the LETKF

```
sbatch run_main.sh
```

The screen output is saved to “screen_output.txt”.

Step 4: Plot

Load the Anaconda environment for Python

```
module load anaconda/2023.09-0/met-env
```

Plot the background error correlation between streamfunction at a selected grid point and streamfunction at all grid points in physical space

```
python plot_ensemble_covariance.py <variable> <ensemble_size> <radius_m>
```

```
python plot_ensemble_covariance.py x 10 5e6
```

Plot truth field with winds and analysis error field with observation locations

```
python draw_letkf_v2.py <analysis_time>
```

```
python draw_letkf_v2.py 2010-01-01T18:00:00Z
```
